AKTU B.Tech Examination

Subject: Design and Analysis of Algorithms

Time: 3 Hours Maximum Marks: 70

Instructions

1. Section A: Attempt all 7 questions. Each carries 2 marks.

2. Section B: Attempt any 3 out of 5 questions. Each carries 7 marks.

3. Section C: Attempt 5 questions. Each has two alternatives; attempt one alternative per question. Each carries 7 marks.

SECTION A — 2 Marks Each

1. Name the sorting algorithm that is most practically used and also write its Time Complexity

2. Differentiate between Backtracking and Branch and Bound Techniques

3. Solve the recurrence T (n) = 4T(n/2) + n2 2

4. Explain Single source shortest path

5. What are approximation algorithms? meant by P (n) approximation algorithms?

6. Prove that if n>=1, then for any n-key B-Tree of height h and minimum degree t >~2, h<= log « ((n +1)/2)

7. Define principal of optimality. When and how dynamic programming is applicable

SECTION B — 7 Marks Each

1. What is Minimum Cost Spanning Tree? Explain Kruskal's Algorithm and Find MST of the Graph. Also write its Time-Complexity

2. Explain HEAP-SORT on the array. Illustrate the operation of HEAP-SORT on the array A= {6, 14, 3, 25, 2, 10, 20, 7, 6}

3. What is dynamic programming? How, is this approach different from recursion? Explain with example. Fa

4. Show the results of inserting the keys F, S, Q, K, C, L, H, T, V, W, M, R, N, P, A, B, X, Y, D, Z, E in order into an empty B-tree. Use t=3, where t is the minimum degree of B- tree. 10

5. Solve the following By Recursion Tree Method T(n)=n+T(n/5)+T(4n/5)

SECTION C — 7 Marks Each

Q1. Attempt either option:

(a) Write an algorithm for insertion of key in the Red-Black Tree. Discuss the various cases for insertion of key in red-black tree for given sequence of key in an empty red-black tree- 5, 16, 22, 25, 2, 10, 18, 30, 50, 12, 1. 10

(b) Explain and write an algorithm for union of two binomial heaps and also write its time complexity? 10 5

Q2. Attempt either option:

(a) Describe in detail Knuth-Morris-Pratt string ching algorithm. Compute the pref x function 7 for the pattern ababbabbabbababbabb when the alphabet is ~ = {a,b}

(b) What is an approximation algorithm?' ne is meant by P (n) approximation algorithms? Discuss approximation algorithm for Travelling Salesman Problem

Q3. Attempt either option:

(a) Define minimum spanning tree (MST). Write Prim's algorithm to generate a MST for any given weighted graph. Generate MST for the following graph using Prim's algorithm. 10

(b) Explain Dijkstra's algorithm to solve single source shortest path problem with suitable example. 10 6

Q4. Attempt either option:

(a) Among Merge sort, Insertion sort and quick sort which sorting technique is the best in worst case. Apply the best one among these algorithms to Sort the list E, X, A, M, P, L, E in alphabetic order

(b) Solve the recurrence using recursion tree method: T (n)=T (1/2) + T (W/4) + T (1/8) +0 Sub Code: RCSS02 Paper Id: 110502 Roll No: L | | | | | | | 4

Q5. Attempt either option:

(a) What is the application of Fast Fourier Transform (FFT)? Also write the recursive algorithm for FFT

(b) Give a linear time algorithin to determine if a text T is a cycle rotation of another string T* For example, 'RAJA' and 'JARA' are cyclic rotations of each other